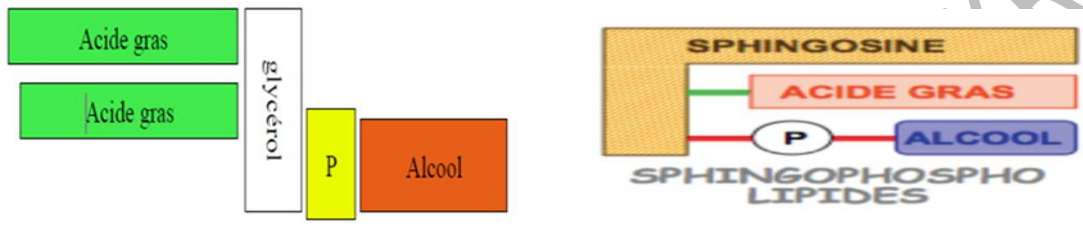


PHOSPHOLIPID METABOLISM

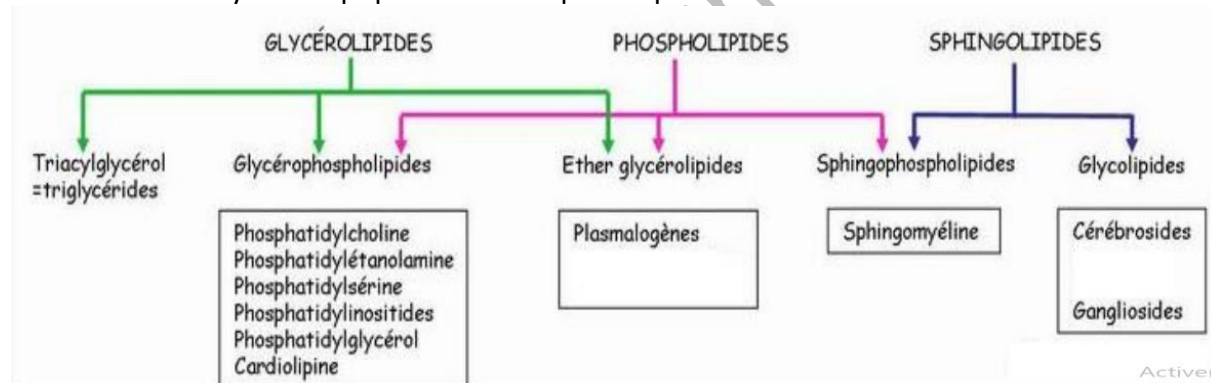
I/Introduction

Phospholipids are esters of fatty acids and an alcohol, the latter being linked to another alcohol via a phosphoryl group.

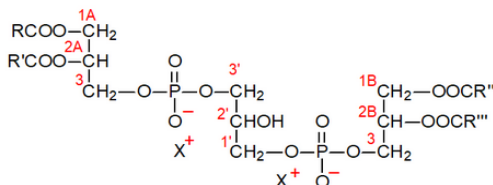


Structural lipids:

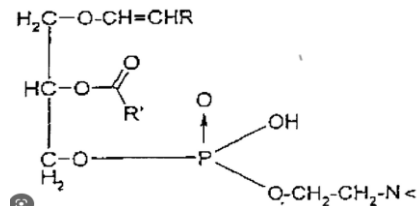
- Biological membranes
- Surface monolayers of lipoproteins and lipid droplets



Structure of cardiolipin



structure of plasmalogen



a) The metabolism of glycerophospholipids (GP) involves:

- ✓ Their synthesis from 1,2-diglyceride and alcohol, both of which must first be activated.
- ✓ Their breakdown by phospholipases.

b) The metabolism of sphingomyelins comprises:

- Their synthesis from palmitoyl-coenzyme A and serine.
- Their catabolism by specific enzymes.

- ❖ **The importance of this metabolism:** The metabolic role of certain GP is as precursors for the synthesis of molecules of biological interest:

-IP3 and DAG

-Polyunsaturated fatty acids, precursors of prostaglandins

-Fatty acids at position 2 of lecithin in the esterification of cholesterol

- ❖ **Phosphatidic acid** lies at the crossroads of triglyceride and GP synthesis.

It is derived mainly from the double acylation of glycerol-3-phosphate, which originates from:

- either **glycerol**, in the liver,
- or **dihydroxyacetone phosphate**.

II/The synthesis of glycerophospholipids (GPLs)

The substrates for GPL synthesis are: 1,2-diglyceride and an alcohol that has been previously activated in the cytidyl form: **cytidine diphosphate (CDP)** provides the phosphate group. Two synthetic pathways are possible:

- The CDP-DG pathway: leading to non-nitrogenous GPs
- The CDP-Alcohol pathway: leading to nitrogenous GPs

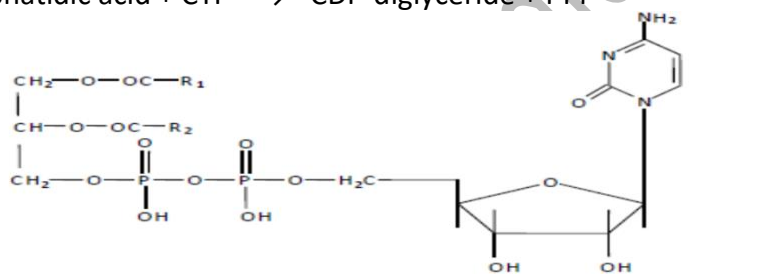
Both pathways are located in the endoplasmic reticulum.

1 - CDP-diglyceride pathway:

a) **Phosphatidic acid** is condensed with CTP to produce cytidine diphosphate diglyceride, the common precursor of **all phosphoglycerides**

Enzyme: **specific transferase**

Phosphatidic acid + CTP $\rightarrow$ CDP-diglyceride + PPi



Cytidine diphosphate di-glyceride

b) **CDP diglyceride** can be regarded as a carrier of phosphatidic acid for the biosynthesis of various phospholipids by reacting with inositol, glycerol phosphate and serine:

Enzyme: **specific synthase**

CDP diglyceride + Inositol $\rightarrow$ **Phosphatidyl-inositol** + CMP

CDP Diglyceride + Serine $\rightarrow$ **Phosphatidylserine** + CMP

CDP Diglyceride + Glycerol phosphate $\rightarrow$ **Phosphatidylglycerol phosphate** + CMP

C) Each of these glycerophospholipids can give rise to other GP;

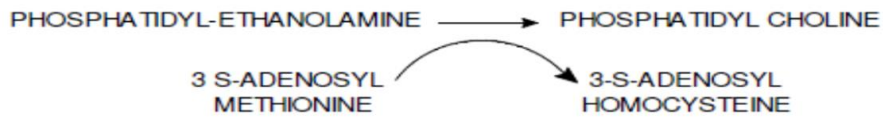
C1- Phosphatidyl inositol is the precursor of two derivatives:

- phosphatidyl inositol monophosphate
- phosphatidyl inositol diphosphate

C2- Decarboxylation of the serine residue in phosphatidyl serine yields phosphatidyl ethanolamine.

Phosphatidylserine $\rightarrow$ **Phosphatidylethanolamine** + CO₂

Phosphatidylethanolamine is itself the precursor of **phosphatidylcholine**: successive transfer of three methyl groups, borrowed from **three molecules of S-adenosylmethionine**



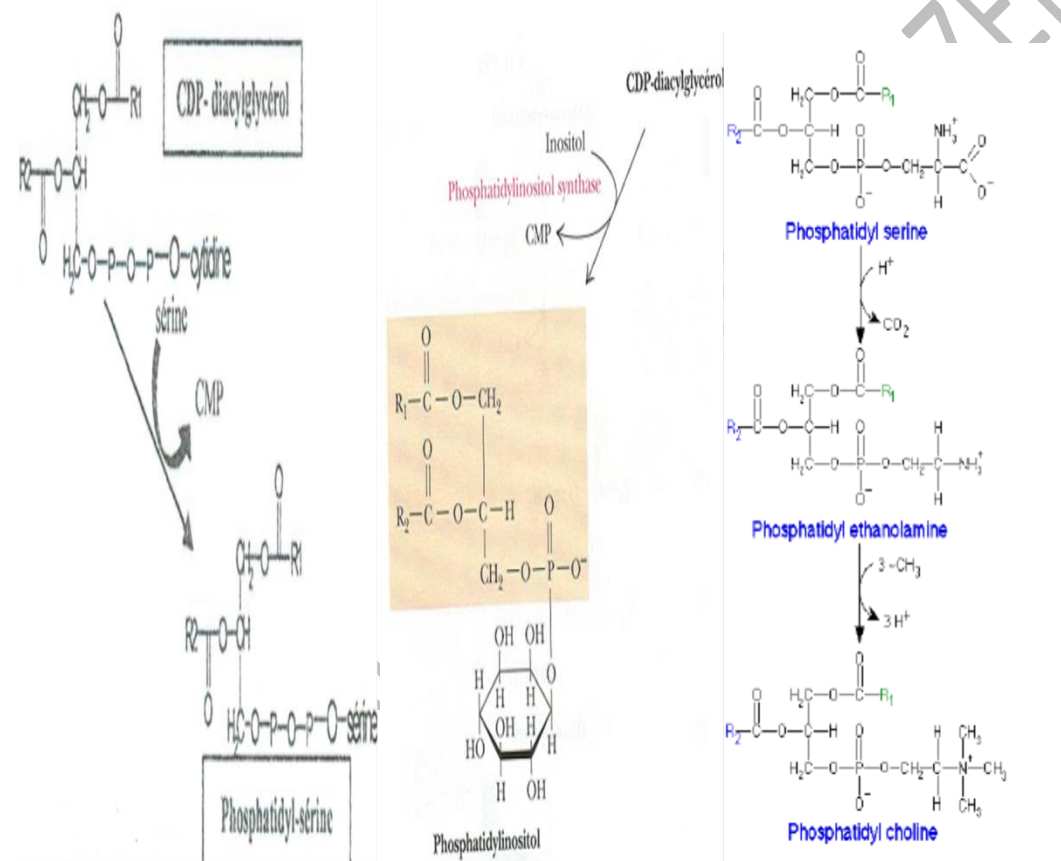
C3 – Phosphatidylglycerol phosphate leads to cardiolipin

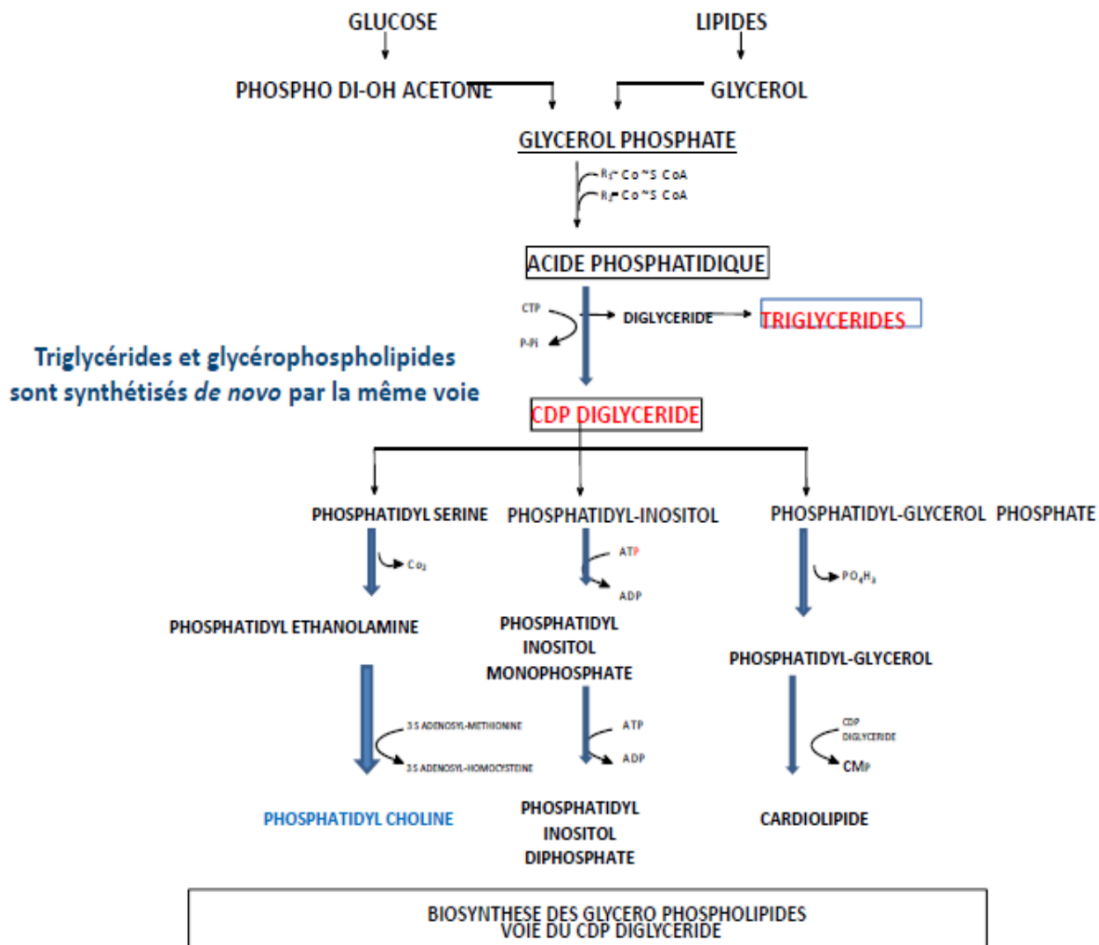
1. Formation of phosphatidylglycerol (under the action of a phosphatase)

Phosphatidylglycerol phosphate $\rightarrow$ **phosphatidyl glycerol** + phosphate

2 - Condensation with a CDP diglyceride molecule

Phosphatidylglycerol + CDP diglyceride $\rightarrow$ **cardiolipin** + CMP





2 - CDP- Alcohol pathway

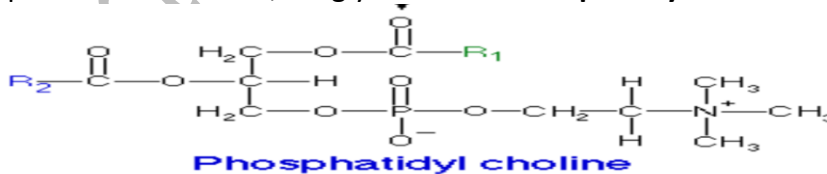
- ❖ This pathway allows for the direct use of **Alcohol** [from dietary intake or the breakdown of endogenous phospholipids (recovery)]

-Exp1: Choline

Step 1: Choline + ATP → **Phosphorylcholine** + ADP

Step 2: CTP + Phosphorylcholine → **CDP-Choline** + PP

Step 3: CDP-Choline + 1,2-Diglyceride → **Phosphatidylcholine** + CMP

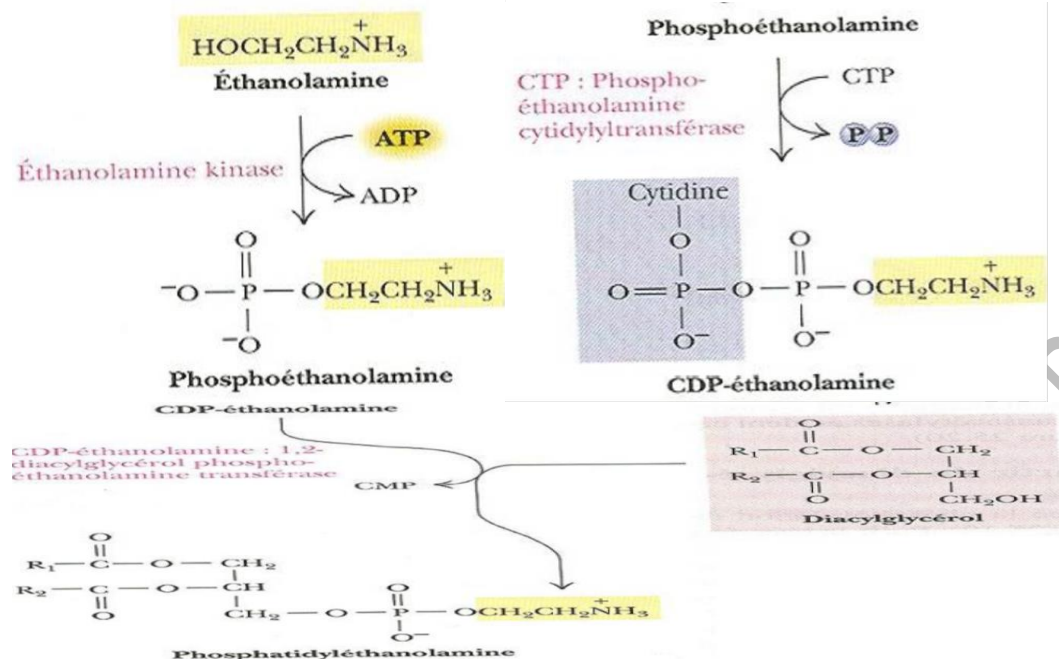


-Very similar reactions lead to cephalins (phosphatidylethanolamine)

1/ Ethanolamine + ATP → **PhosphorylEthanolamine** + ADP

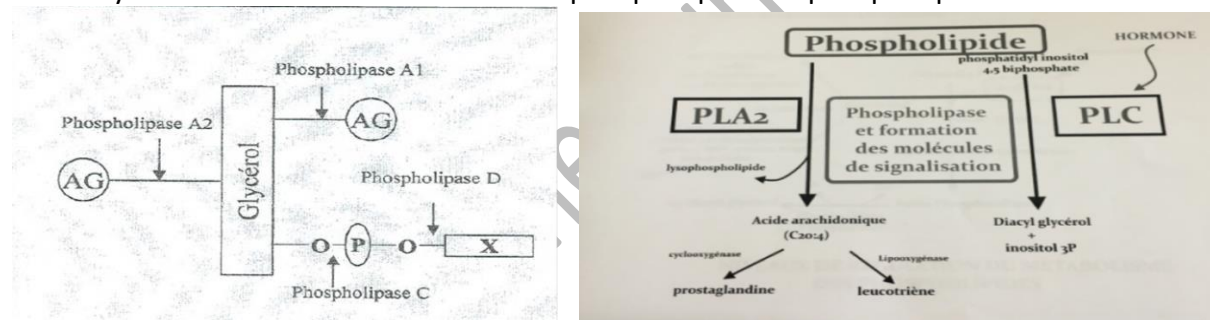
2/CTP+ Phosphoryl-Ethanolamine → **CDP-Ethanolamine** + PPi

3/CDP-Ethanolamine+1,2Diglyceride → **Phosphatidyl-Ethanolamine** + CMP



III/Degradation of phospholipids

The enzymes involved in the catabolism of phospholipids are phospholipases. These include:



Phospholipases and the formation of signaling molecules:

IV/ The synthesis of sphingolipids

A/ Introduction

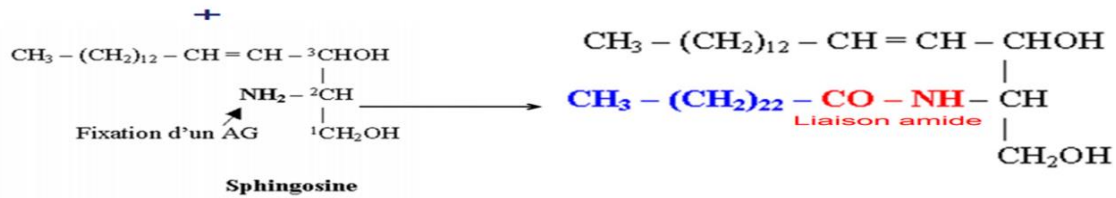
Sphingomyelins: Phospholipids

- Gangliosides
 - Cerebrosides
 - Sulfatides
- } Glycolipids

Sphingolipid precursor molecule: **Ceramide**: an amide of fatty acid (FA) and alcohol,

- Alcohol: **sphingosine**, a diol amine with a long carbon chain
- The attachment of a fatty acid to the amine group via an **amide bond**

Acide lignocérique (C_{24:0})

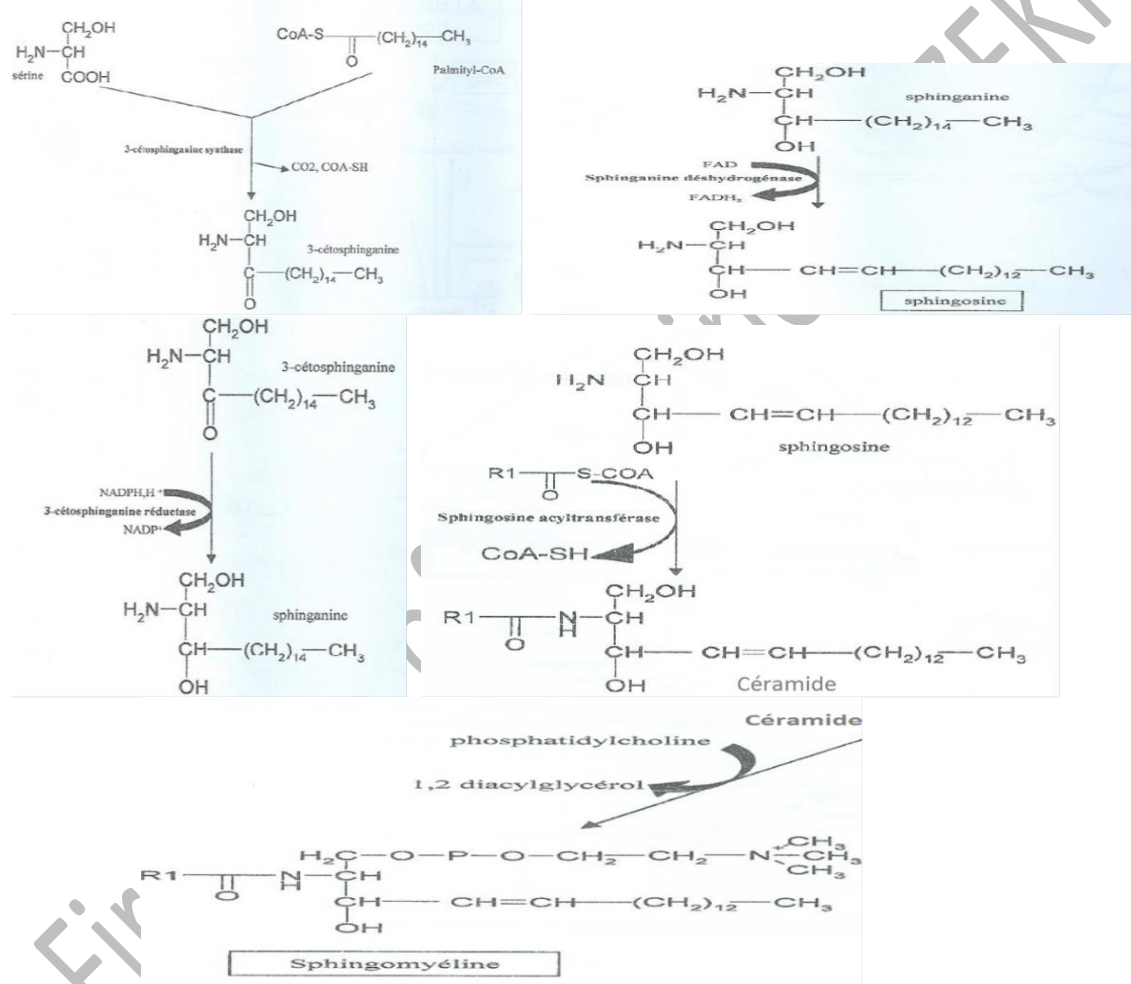


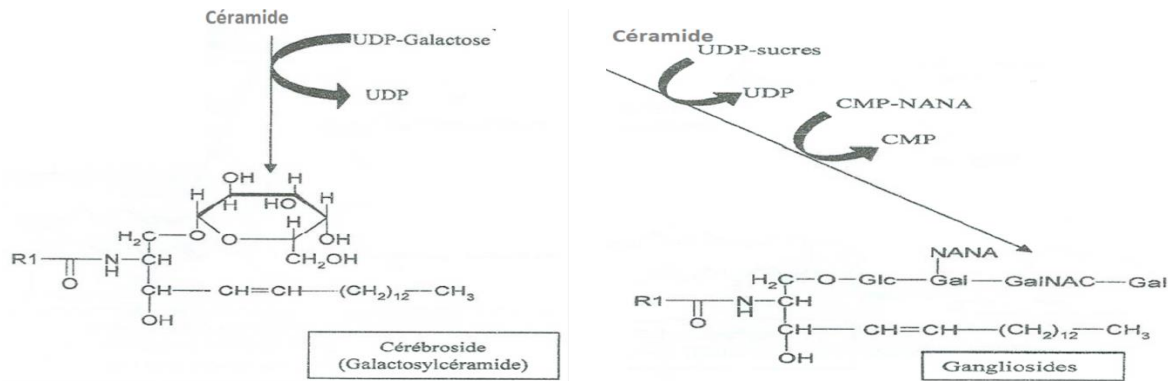
B/ Ceramide synthesis

Ceramide, the leading sphingolipid, is generated in cells via various pathways:

- either by de novo synthesis (using serine and palmitoyl CoA as substrates)
- or by recycling from sphingolipids

General diagram of sphingolipid synthesis:





V/Catabolism of sphingolipids

- Sphingolipids are hydrolysed by lysosomal enzymes:
Various acid hydrolases, specific to the terminal sugars of sphingolipids
– Examples: α -galactosidase, β -galactosidase, β -glucosidase, neuraminidase (sialidase), hexosaminidase, sphingomyelinase, sulfatase (sulfate esterase), and ceramidase.
- The hereditary absence of one of these enzymes: hereditary sphingolipid storage diseases or **sphingolipidoses**:
- Examples of **sphingolipidoses**

Disease	Deficient enzyme	Accumulated substance
Gaucher disease	β -glucosidase	Glucocerebroside
Niemann-Pick disease	Sphingomyelinase	Sphingomyelin
Tay-Sachs disease	Hexosaminidase A	GM2 ganglioside